Universiti Tun Hussein Onn Malaysia

UNIVERSITI TUN HUSSEIN ONN MALAYSIA

FINAL EXAMINATION
SEMESTER I
SESSION 2023/2024

COURSE NAME	:	OPERATING SYSTEM
COURSE CODE	:	BIC 20803
PROGRAMME CODE	:	BIS / BIP / BIW / BIM
EXAMINATION DATE	:	JANUARY / FEBRUARY 2024
DURATION	:	3 HOURS
INSTRUCTIONS	:	1. ANSWER ALL QUESTIONS 2. THIS FINAL EXAMINATION IS CONDUCTED VIA ☐ Open book ☒ Closed book 3. STUDENTS ARE **PROHIBITED** TO CONSULT THEIR OWN MATERIAL OR ANY EXTERNAL RESOURCES DURING THE EXAMINATION CONDUCTED VIA CLOSED BOOK

THIS QUESTION PAPER CONSISTS OF **FIVE (5)** PAGES

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Q1 Based on the given scenario, answer **Q1(a)-Q1(e).**

A busy hospital emergency room (ER) handles a large number of patients with varying levels of severity in their conditions that require a priority level (1 is the highest). To ensure efficient and timely care, the hospital employs an electronic patient tracking and scheduling system that uses the Shortest Remaining Time First scheduling algorithm. Table Q1.1 shows the queue of patients and treatment hours required with priority for today's operation.

Table Q1.1

Patient	Time Arrive	Treatment Hour	Priority
Hasan	00:00	12	4
Aminah	06:00	1	1
Ebrahim	07:00	6	3
Elyas	10:00	2	2

(a) Identify sequence of the treatment process.

(6 marks)

(b) Assuming that during the treatment of Elyas, Daniel comes at 11:00 with the highest priority that requires one hour of treatment.
Identify the new sequence of the treatment process.

(2 marks)

(c) Describe **TWO (2)** benefits of applying Shortest Remaining Time First scheduling algorithm in a hospital emergency room.

(4 marks)

(d) Find the average waiting time and average turnaround time if pre-emptive Priority scheduling algorithm is applied to the queue in **Table Q1.1**. Show your calculations.

(4 marks)

(e) Based on your answer in **Q1(d)**, analyse **TWO (2)** disadvantages of context switch.

(4 marks)

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Q2 Based on Resource Allocation Graph in **Figure Q2.1**, determine whether deadlock exist or not. Justify your answer.

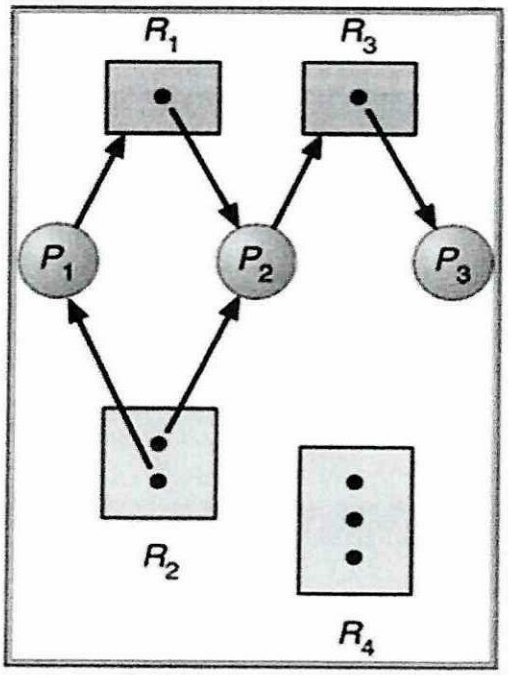


Figure Q2.1

(4 marks)

Q3 Based on the given scenario, answer the following questions.

An online booking ticket system is running on a server. The system options are to make reservation, inquiry about the ticket, modify the reservation, cancel the reservation and exit.

(a) Suggest **THREE (3)** sub-tasks of making reservation.

(6 marks)

(b) Based on your answer **Q3(a)**, explain parallelism process.

(6 marks)

(c) Based on your answer **Q3(b)**, draw the parallel process.

(4 marks)

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Q4 Given six memory partitions in order of `100 KB, 500 KB, 200 KB, 300 KB, 600 KB and 700 KB` and five processes as in **Table Q4.1**

Table Q4.1

Process	Size (KB)
P_1	120
P_2	460
P_3	680
P_4	330
P_5	180

(a) Illustrate the new memory partitions after all processes are loaded using the following algorithms:

(i) First fit.

(5 marks)

(ii) Best fit.

(5 marks)

(iii) Worst fit

(5 marks)

(b) Based on your answer in **Q4(a)**, analyze which of these algorithms might lead to the most internal fragmentation after the allocation for this scenario.

(5 marks)

Q5 Consider page reference string `m, k, m, r, k, q, m, r, p, q.`

(a) Perform the access sequence for the case of a cache capacity of 3 pages using the following algorithms:

(i) First-In-First-Out (FIFO) page replacement algorithm.

(5 marks)

(ii) Least Recently Used (LRU) page replacement algorithm.

(5 marks)

(b) Based on your answers in **Q5(a)**, calculate the hit rate and the miss rate for the FIFO and LRU algorithms.

(4 marks)

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(c) Compare and contrast the Optimal Page Replacement Algorithm with LRU Algorithm, focusing on:

(i) Decision criteria for page replacement.

(2 marks)

(ii) Page fault rates.

(2 marks)

(iii) Practical implementation.

(2 marks)

Q6 **Figure Q6.1** shows a set of blocks of a secondary storage. Answer **Q6(a)-Q6(c).**

	F1	F1			
F2	F2			F3	F3
F3	F3	F3			
	F4	F4	F4		
		F5	F5	F5	

Figure Q6.1

(a) What is the bit vector value of the disk blocks?

(4 marks)

(b) Identify the directory entry of the disk blocks.

(5 marks)

(c) Using the best-fit allocation strategy for contiguous storage, identify the new directory entry for disk blocks when a file named **F6** with a size of 56KB is added, assume that each block size is 37KB.

(6 marks)

Q7 Identify the importance of security measures in an operating system and list **THREE(3)** potential risks must the system defend against.

(5 marks)

- END OF QUESTIONS -

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